

Prakhar Shekhar Parthasarathi

Full-Stack Software Engineer / Java & Spring Boot Backend / React Front-End / AWS

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🐙 github.com/praxstack · 🌐 linkedin.com/in/prakharshekhar

47%

P90 LATENCY CUT ·
BOOKING SURFACES

+33%

CONVERSION LIFT

195k rps

GET · JAVA REDIS FROM
SCRATCH

20→5

ON-CALL TICKETS /
SHIFT

≡ SUMMARY

Full-stack software engineer with **3+ years at Amazon** across Payments and Travel — **owned 4 microservices end-to-end** through design, build, deploy, and on-call. Java-heavy backend (**Java 17 / Spring Boot**, JVM concurrency with `CompletableFuture`, bounded `ExecutorService`, `ConcurrentHashMap`), idempotent event-driven architecture over **SQS + Step Functions**, and typed infrastructure in **AWS CDK (Java)** — paired with **React (SSR & CSR) + Redux** for customer-facing flows. Customer-facing impact: **47% p90** reduction on booking surfaces, **33% conversion** lift, **43 bps CPT** reduction via idempotent schedule-change automation, on-call tickets cut **20+ to ~5** per shift.

⌘ CORE TECHNICAL STACK

🔧 JAVA & JVM

Java 17, Spring Boot, Spring MVC, Spring DI, Google Guice, JUnit 5, Mockito, SLF4J, Lombok, Maven, Gradle · `ExecutorService`, `CompletableFuture`, `ConcurrentHashMap`, thread-pool tuning, JVM profiling, GC analysis

☁ AWS

Lambda (Java), DynamoDB, SQS, Step Functions, API Gateway, AppSync, CloudWatch, ECS, **CDK (Java)** — typed IaC for all service infrastructure

🗄 DATA & STORAGE

DynamoDB, Redis, NoSQL modelling, SQL, caching strategies

☀ TOOLING & PROCESS

Git, GitHub, CI/CD, Gradle JaCoCo coverage gates, Agile / Scrum, on-call, code review, incident response

🔗 DISTRIBUTED SYSTEMS

Microservices, REST, GraphQL, event-driven architecture, idempotency & dedup keys, state machines, retry with exponential backoff, DLQ handling, at-least-once semantics, TDD

📱 FRONT-END

React.js (SSR & CSR), Redux (1.5 yrs production) — customer-facing Amazon flows; HTML / CSS, JavaScript / TypeScript

📄 OTHER LANGUAGES

Python, C++, VTL

Software Development Engineer

Jul 2022 – Sep 2025

📍 Amazon · Travel (Hotels & Flights) · Bengaluru

- **Service ownership — Spring Boot, Spring MVC.** Owned 4 **customer-facing microservices** powering the hotel booking platform — search, listing, room selection, order review — across design, code review, deploy, and on-call. Built REST controllers, service-layer logic, and `@Repository` DynamoDB DAOs wired through Spring DI. Shipped Prime-exclusive discount flow and partner-discount integration with tuned cache logic — **~30 bps** drop in funnel churn.
- **JVM concurrency — CompletableFuture, bounded executors.** Refactored sequential downstream calls into `CompletableFuture.allOf()` on a bounded `ExecutorService`; parallelised 6 partner-API lookups — hotel landing p90 cut **47%** (**1.1s → 0.59s**), conversion up **+33%**. GZIP on 150–200 MB catalog payloads cut wire size **95%**, eliminating network-timeout failures.
- **Event-driven — SQS + Step Functions.** Designed idempotent Java webhook for real-time flight schedule changes (cancellations, date-changes, non-operational flights) — SQS ingestion, DynamoDB-backed dedup keys, audit logging, automated offline modifications — addressing the highest-impact customer-pain category (**43 bps CPT**). Orchestrated end-to-end booking state machine (confirm / cancel / retry) on Step Functions with Java Lambda handlers.
- **Infrastructure as code — AWS CDK (Java).** Authored all service IaC in **Java CDK** — Lambdas, DynamoDB, SQS, Step Functions, API Gateway, CloudWatch alarms. Migrated 2 services from hand-rolled CloudFormation to typed CDK stacks — compile-time infrastructure checks, **halved** rollback incidents.
- **Testing — JUnit 5, Mockito, Spring Boot.** Held **80%+** line coverage; Mockito argument captors for downstream contract verification; `@SpringBootTest` integration tests against in-memory DynamoDB. Introduced Gradle JaCoCo coverage gates in CI.
- **Operational excellence & high-bar automation.** Dove deep into multi-year on-call corpus — tickets cut from **20+ to ~5 per shift** (**~2 SDE-weeks / week** reclaimed). Java email-automation service deprecated stale WebLab experiments — config-hygiene incidents from 5–6 to **0** per on-call. Resolved **150+** production incidents; authored COEs and drove action items to closure. Built auto-insurance purchase flow (Java + React) with milestone persistence and auto-fill — **60%+** lift in renewals.

Software Development Engineer, Intern

May 2021 – Jul 2021 · Feb 2022 – Jul 2022

📍 Amazon · Bengaluru

- Migrated legacy GraphQL resolvers to **Java** — rebuilt resolver classes, wired them to DynamoDB via Spring-injected DAOs, exposed through API Gateway + Lambda.
- Shipped **One-Click Renewal** for auto-insurance using past-purchase data — async Java pre-fill workflow that materially simplified the funnel.
- Applied OO design and **Google Guice DI** throughout; landed a clean, testable codebase with **90%+** unit-test coverage (JUnit 5 + Mockito).

SELECTED PROJECTS

Redis Server in Java

github.com/praxstack/redis-server-java

Java 17

Maven

JUnit 5

Mockito

ConcurrentHashMap

ScheduledExecutorService

RESP2-compatible Redis server built from scratch. Thread-per-client on a bounded `ExecutorService`; lock-free reads via `ConcurrentHashMap`; hybrid TTL eviction (lazy check on `GET` plus active `ScheduledExecutor` sweep); graceful shutdown via `Runtime.addShutdownHook`; integration tests over real TCP sockets.

Benchmarks (redis-benchmark, 100 concurrent clients):
SET **154k rps** · GET **195k rps** · INCR **128k rps**
· p50 < **0.3 ms** · 42 tests, 100% pass.

Markdown Viewer App

github.com/praxstack/markdown-viewer-app

Vite

Vanilla JS

Marked.js

PrismJS

Mermaid

Vitest

Production-ready markdown viewer with real-time preview, syntax highlighting, Mermaid diagram rendering, 10 professional themes, and PDF / HTML export. Architected with modular vanilla JavaScript — zero framework overhead, sub-100 KB bundle.

Metrics: **155+** Vitest tests · real-time preview · Mermaid diagrams · PDF / HTML export.

EDUCATION

M.C.A., Computer Science · Motilal Nehru National Institute of Technology (MNNIT), Prayagraj **GPA 9.25** **2019 – 2022**

B.Sc. (Hons), Computer Science · Bhaskaracharya College of Applied Sciences, University of Delhi **2015 – 2018**

AWARDS & RECOGNITION

 **Amazon Pay Merchants Categories Champion** — 100+ hotel trust-buster bugs resolved; on-call tickets cut **20+ to ~5** per shift. **2023**